



been established. The final decision involves finding a balance between the required properties and cost-effectiveness to achieve the desired battery performance.

Rules for better escape

Regulations governing EVs are in a state of flux, struggling to keep pace with rapid technological advancements. A key example is the regulation surrounding thermal runaway, a potential safety hazard in EV batteries.

China was the pioneer in implementing specific regulations for thermal runaway, requiring vehicles to allow five minutes from the start of a thermal event for occupants to safely exit the vehicle. This standard became a benchmark for many regions.

However, as technology progresses, so do safety standards. Future regulations in Europe and the US are expected to become even more stringent, possibly requiring escape times of 10, 15, or even 30 minutes. This shift is driven by the recognition that in certain scenarios, such as being trapped in a tunnel, five minutes might not be sufficient for a safe escape.

The initial adoption of these regulations in China led to increased use of materials that could provide longer protection times. Now, as other regions move towards similar standards, original equipment manufacturers (OEMs) are focusing on providing materials that meet these extended safety requirements.

It is important to note that while

Green focus in EV fire protection

Environmental and sustainability considerations are increasingly influencing the choice of fire protection materials in EVs. As EVs are viewed as a green technology, there is a growing focus on the sustainability of the materials used in their construction.

Sourcing of materials. The sustainability of materials like mica, used for fire protection, is under scrutiny due to concerns about mining practices. Efforts are being made to ethically and sustainably source these materials, such as the Responsible Mica Initiative.

Recyclability. Most fire protection materials are not recyclable, which poses a challenge for sustainability. Recycling is currently focused on valuable components like nickel, cobalt, and lithium, while other materials are often overlooked.

Supplier initiatives. Suppliers are working to enhance the sustainability of their products, but it is difficult to rank materials based on sustainability due to various factors involved. It is also not always the primary decision factor in material selection.

In conclusion, while environmental and sustainability considerations present challenges, they also drive innovation in the development of fire protection materials for EVs. The focus on sustainability is leading to the exploration of new materials and practices that balance safety, performance, and environmental impact.

regulations are driving the adoption of certain materials, they do not specify any particular material. Instead, they focus on ensuring that materials used in EVs allow for the required escape times in case of a thermal event. This approach promotes innovation and the adoption of safer materials across the industry.

The market for fire protection materials in EVs is currently dominated by established options, such as ceramic blankets, mica sheets, and encapsulating foams. These materials are expected to continue playing a significant role in the industry. However, as we look towards the future, there is a clear trend towards the adoption of more innovative alternatives, such as aerogel materials and compression pads with fire protective additives. These emerging solutions are gaining traction within

vehicles, offering enhanced performance and safety.

The growing EV market, particularly in China, which holds the largest market share, is set to drive a substantial increase in demand for fire protection materials. This growth is not limited to China; it is a global trend, with significant activity from companies worldwide, including those in Asia, Europe, and the US. As the industry evolves, the focus will remain on finding the most effective and sustainable solutions to ensure the safety and reliability of EVs, with China leading the demand due to its vast vehicle market. **EFY**

The Co-author, Nidhi Agarwal is a journalist at EFY. She is an Electronics and Communication Engineer with over five years of academic experience. Her expertise lies in working with development boards and IoT cloud. She enjoys writing as it enables her to share her knowledge and insights related to electronics with like-minded techies.

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